

UC Berkeley Math 128A: Homework 2

Prof. Per-Olof Persson (persson@berkeley.edu)

MATLAB usage classification: PEN (pen and paper only), CALC (use as calculator), CODE (run codes).

0. Please assign yourself a self grade on this homework (0, 1, or 2) based off the guidelines in the syllabus.
1. (1.3-1a, CODE) Use three-digit chopping arithmetic to compute the sum $\sum_{i=1}^{10} (1/i^2)$ first by $\frac{1}{1} + \frac{1}{4} + \dots + \frac{1}{100}$ and then by $\frac{1}{100} + \frac{1}{81} + \dots + \frac{1}{1}$. Which method is more accurate, and why?
2. (1.3-6, PEN) Find the rates of convergence of the following sequences as $n \rightarrow \infty$
 - a. $\lim_{n \rightarrow \infty} \sin \frac{1}{n} = 0$
 - b. $\lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0$
 - c. $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n}\right)^2 = 0$
 - d. $\lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)] = 0$
3. (1.3-7, PEN) Find the rates of convergence of the following functions as $h \rightarrow 0$
 - a. $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$
 - b. $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0$
 - c. $\lim_{h \rightarrow 0} \frac{\sin h - h \cos h}{h} = 0$
 - d. $\lim_{h \rightarrow 0} \frac{1 - e^h}{h} = -1$
4. (1.3-11, PEN) The sequence $\{F_n\}$ described by $F_0 = 1, F_1 = 1$, and $F_{n+2} = F_n + F_{n+1}$, if $n \geq 0$, is called a Fibonacci sequence. Its terms occur naturally in many botanical species, particularly those with petals or scales arranged in the form of a logarithmic spiral. Consider the sequence $\{x_n\}$, where $x_n = F_{n+1}/F_n$. Assuming that $\lim_{n \rightarrow \infty} x_n = x$ exists, show that $x = (1 + \sqrt{5})/2$. This number is called the golden ratio.
5. (2.1-1, CALC) Use the Bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x$ on $[0, 1]$.
6. (2.1-12, CODE) Let $f(x) = (x+2)(x+1)^2x(x-1)^3(x-2)$. To which zero of f does the Bisection method converge when applied on the following intervals?
 - a. $[-1.5, 2.5]$
 - b. $[-0.5, 2.4]$
 - c. $[-0.5, 3]$
 - d. $[-3, -0.5]$
7. (2.1-17, CALC) Use Theorem 2.1 to find a bound for the number of iterations needed to achieve an approximation with accuracy 10^{-4} to the solution of $x^3 - x - 1 = 0$ lying in the interval $[1, 2]$. Find an approximation to the root with this degree of accuracy.